03. Celestial amplitudes: distributional crossing and soft-limit consistency

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Instead of describing a collision by particle energies and momenta, imagine recording the directions in which particles enter or leave on a sphere surrounding the experiment. Celestial methods reorganize scattering data into quantities associated with that sphere. The ambition is a new language for gravity. Before using that language to discover new theories, this problem checks that it faithfully preserves familiar facts about a simple graviton collision.

Key terms

- **Celestial sphere** — The sphere of directions seen from a point, used here to label asymptotic particle directions rather than a physical shell.
- **Graviton** — The hypothetical quantum excitation of the gravitational field used in perturbative quantum-gravity calculations.
- **Helicity** — The component of a particle's spin along its direction of motion.
- **Mellin transform** — A mathematical transformation that reorganizes dependence on energy into dependence on scaling behavior.
- **Conformal weight Δ** — A label describing scaling behavior in the celestial description; it can be complex.
- **Distribution** — A generalized function, such as a delta function, understood through integration against test functions rather than ordinary pointwise values.
- **Crossing** — The relationship between different scattering processes obtained by reinterpreting an incoming particle as an outgoing antiparticle, with appropriate continuation.
- **Soft limit** — The limit in which one particle's energy approaches zero; a soft theorem controls the resulting leading behavior.
- **Regulator** — A temporary modification that makes a singular expression well defined, accompanied by a rule for removing or interpreting it.

Why this matters

A successful reformulation could expose symmetries or constraints that are difficult to see in ordinary scattering variables. That would give theorists another way to investigate gravity. But a change of mathematical language is useful only if singularities, conservation laws and limiting cases survive the translation correctly.

What progress would mean

A verified example would supply a dependable foundation for later celestial calculations. It would establish that specific identities translate correctly, rather than establish a complete two-dimensional theory of the universe.

Core question

Can a regulated celestial transform of a specified four-graviton tree amplitude reproduce Lorentz covariance, crossing, and the leading conformally soft residue in one consistent convention?

Scope and assumptions

Fix four-dimensional Einstein gravity at tree level, a nonzero four-graviton helicity amplitude, incoming/outgoing signs ϵ_i, and $\mathrm{p}_i{=}\epsilon_i\omega_i\mathrm{q}(\mathrm{z}_i,\bar{z}_i)$. Declare metric, helicity, normalization, and analytic-continuation conventions. The target is an identity pipeline for amplitudes, not a classification of celestial CFTs.

Use $\Delta_i{=}1{+}\mathrm{i}\lambda_i$ on the principal series where appropriate and $\mathrm{h}_i{=}(\Delta_i{+}\mathrm{J}_i)/2$, $\bar{h}_i{=}(\Delta_i{-}\mathrm{J}_i)/2$. Identify explicitly which continuations away from that contour are used for soft residues.

Mathematical target

Define the transformed distribution by

$$\widetilde{\mathcal{A}} = \prod_i \int_0^\infty d\omega_i\, \omega_i^{\Delta_i - 1}\; \delta^{(4)}\left(\sum_i \epsilon_i \omega_i q_i\right) \mathcal{M}_4.$$

State a regulator or test-function space making the integral meaningful, and specify regulator removal or meromorphic continuation. From a leading momentum-space term proportional to ω_s^{-1}, derive the Mellin pole at $\Delta_\mathrm{s}{=}1$ in this convention. Subleading residues require their own derivation. Crossing must transform helicities, signs, and support, not just permute ordinary Euclidean functions.

Required outputs and proof obligations

Produce exact momentum-space expressions, regulated transform formulas, support conditions, and symbolic covariance/crossing/soft identities. If a transform cannot be defined under the chosen assumptions, return a precise obstruction at that step. A positive bootstrap is an extension only after deriving an applicable inner product and positivity statement.

Validation and rejection controls

Check the original amplitude's factorization first. Test Mellin formulas against analytic one-variable examples in their convergence strips. Verify soft residues before and after transformation under stated limit-interchange hypotheses. Use smeared distributions for numerical checks; never evaluate a delta distribution as an ordinary function.

Milestones and research extension

Baseline: one four-point transform with explicit support and checked leading soft residue. Strong: a second helicity/process example and a verified crossing relation. Research extension: derive a justified positive constraint in a restricted subsector before attempting an SDP island.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Adamo et al., Celestial amplitudes and conformal soft theorems — Conformally soft graviton operators at Δ=1 and Δ=0.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$p_i = \epsilon_i \omega_i q_i, \quad \Delta_i = 1 + i\lambda_i, \quad h_i = \frac{\Delta_i + J_i}{2}, \quad \bar{h}_i = \frac{\Delta_i - J_i}{2}.$$

$$\widetilde{\mathcal{A}} = \left(\prod_{i=1}^{4} \int_0^\infty d\omega_i \, \omega_i^{\Delta_i - 1} \right) \delta^{(4)} \left(\sum_i \epsilon_i \omega_i q_i \right) \mathcal{M}_4.$$

The expression is a regulated distribution. A leading soft term proportional to ω_s^{-1} yields the local integral $\int_0^\Lambda \omega_s^{\Delta_s - 2} d\omega_s = \Lambda^{\Delta_s - 1}/(\Delta_s - 1)$ before continuation, exposing the pole at $\Delta_s = 1$. Support, helicity, crossing and regulator removal remain proof obligations; ordinary Euclidean OPE positivity is not assumed.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — missing distributional specification (36–45, 94–113).** A Mellin transform of a stripped amplitude without momentum conservation and convergence rules is incomplete. The numerical quadrature sketch does not define the multidimensional distribution that is actually needed.
2. **Blocking — imported positivity (18–23, 57–75).** Principal-series celestial data cannot simply inherit the positive OPE-coefficient squares of a unitary Euclidean two-dimensional CFT. An SDP needs a proved positivity structure, not an asserted positive norm.
3. **Major — soft poles (60–64).** The generic $\Delta \to 0$ double-pole formula does not reproduce the leading graviton soft pole in the stated Mellin convention. Deriving the pole from the energy power prevents convention mistakes.
4. **Major — scope (15–28, 79–86).** "Consistent space" and "islands" assume tools whose foundational prerequisites are themselves research questions. A distributionally correct identity pipeline is a meaningful prerequisite and preserves the original direction.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Can a regulated celestial transform of a specified four-graviton tree amplitude reproduce Lorentz covariance, crossing, and the leading conformally soft residue in one consistent convention?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: one four-point transform with explicit support and checked leading soft residue. Strong: a second helicity/process example and a verified crossing relation. Research extension: derive a justified positive constraint in a restricted subsector before attempting an SDP island.

Sources supporting the corrections

- Adamo et al., Celestial amplitudes and conformal soft theorems — Conformally soft graviton operators at $\Delta=1$ and $\Delta=0$.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.